Lamport Logical Timestamp Synchronization Engine in php: Causal Clock Verification and Execution Dynamics

MINI PAPER FROM PERSONAL GITHUB LABS

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Abstract—Lamport logical clocks are a cornerstone for establishing causal order in distributed systems, yet practical implementations in high-level scripting languages remain under-explored. This paper presents a PHP-based Lamport Logical Timestamp Synchronization Engine that integrates a height-balanced search index for event logging. We formalize three research questions concerning (RQ1) the asymptotic complexity of timestamp-driven insertions, (RQ2) the preservation of a strict monotonicity invariant across remote updates, and (RQ3) the empirical throughput and latency profile on a modern multi-core workstation. The engine is built from verified source files (engine.php, types.php, main.php) and evaluated over 30 benchmark iterations on a 13th Gen Intel Core i9-13900HX (24 cores/32 threads, 63.8 GB RAM, Windows 11). Measured mean latency is 60 300.89 µs with a throughput of 16 584 ops/s. Results confirm that the balanced index maintains $O(\log n)$ insertion cost while respecting Lamport's causal ordering. .

Source Code: https://github.com/myonathanlinkedin/php-backend-core/tree/main/algorithms/20261011_094516_lamport_logical_timestamp_sync

Index Terms—Lamport clock, causal ordering, balanced search index, PHP runtime, distributed synchronization

1 INTRODUCTION

Distributed applications such as sensor networks, micro-service fabrics, and collaborative editors rely on a consistent view of event causality. Lamport's logical timestamps provide a lightweight mechanism to order events without requiring synchronized physical clocks, but the practical integration of these timestamps with efficient indexing structures is rarely addressed in high-level languages. In industrial settings, PHP back-ends are increasingly employed for real-time coordination, yet their managed runtime introduces non-deterministic latency and garbage-collection pauses that can jeopardize causal guarantees. This investigation addresses three research questions:

1) **RQ1:** What is the asymptotic time complexity of inserting a timestamped event into the engine's balanced search index?
2) **RQ2:** Does the engine preserve the strict monotonicity invariant $L(e_i) < L(e_j)$ for any causally related events $e_i \to e_j$ under concurrent remote updates?
3) **RQ3:** How does the measured throughput and latency distribution compare to the theoretical $O(\log n)$ bound on a contemporary multi-core workstation?

To answer these questions we (i) design a height-balanced binary search tree that stores `LamportTimestamp` objects, (ii) embed the tree within a PHP class that updates timestamps according to Lamport's rules, and (iii) execute a micro-benchmark suite that records wall-clock latency and operations per second. The remainder of the paper is organized as follows: Section 2 surveys prior formalizations of logical clocks; Section 3 details the engine's state-transition model and index layout; Section 4 states and proves the core invariants; Section 5 describes the experimental platform; Section 6 presents empirical findings; Section 7 discusses validity threats; and Section 8 concludes.

2 RELATED WORK

Lamport's original logical clock formulation established the foundation for causal ordering in distributed systems [1]. Subsequent work extended the model to secure authentication using timestamp-based digital signatures, demonstrating feasibility in wireless sensor networks [2]. More recent analysis identified a "logical timestamp skew" anomaly in transaction schedulers that can break monotonicity under high contention [3]. While these studies provide theoretical and security-oriented perspectives, none examine the interaction between Lamport clocks and balanced indexing structures within a PHP runtime. Our contribution bridges this gap by coupling a height-balanced search tree with Lamport's update rules and empirically validating the combined system.

3 LAMPORT LOGICAL TIMESTAMP SYNCHRONIZATION ENGINE: STATE TRANSITION GRAPH AND BALANCED INDEX STRUCTURE

The engine consists of three cooperating components: (i) `LamportClock`, which maintains a mutable `LamportTimestamp`; (ii) `Process`, representing a logical participant that records local events and exchanges `Message` objects; and (iii) `TimestampIndex`, a self-balancing binary search tree (AVL variant) that stores `EventLogEntry` objects ordered by their timestamp value. Upon a local event, `LamportClock::tick()` increments the timestamp and inserts the new entry into the index, triggering a standard AVL rotation if the height-balance factor exceeds one. When a remote message arrives, `LamportClock::updateOnReceive()` computes the maximum of the local and received timestamps, increments the result, and inserts the updated entry. The index guarantees $O(\log n)$ search, insertion, and deletion, where n is the number of logged events. Memory layout follows PHP's native object model; each node contains a `LamportTimestamp`, left/right child references, and a height field. The engine's public API (see `main.php`) exposes `sendMessage` and `receiveMessage` methods that encapsulate the state-transition logic.

4 FORMAL INVARIANTS & PROOFS

We formalize two invariants that underpin correctness.

Invariant 1 (Monotonicity). For any two events `e_i` and `e_j`, if `e_i` causally precedes `e_j` (denoted $e_i \to e_j$), then $L(e_i) < L(e_j)$, where $L(\cdot)$ denotes the associated `LamportTimestamp` value.

$$\forall e_i, e_j \; (e_i \to e_j) \implies L(e_i) < L(e_j) \tag{1}$$

Invariant 2 (Height Balance). After each insertion into `TimestampIndex`, the height difference between left and right subtrees of any node is at most one.

Proof: Monotonicity follows directly from Lamport's update rules. A local event invokes `tick()`, which returns old $+1$, guaranteeing $L(e_{\text{new}}) = L(e_{\text{old}}) + 1$. For a remote receipt, the engine computes $\max(L_{\text{local}}, L_{\text{remote}})$ and then increments, yielding $L(e_{\text{recv}}) = \max(L_{\text{local}}, L_{\text{remote}}) + 1$. In both cases the resulting timestamp exceeds any timestamp that could have causally preceded the event, establishing the strict inequality required by Invariant 1.

Height balance is maintained by the AVL rebalancing procedure. After insertion, the algorithm updates node heights bottom-up and performs at most two rotations (single or double) whenever a balance factor of 2 or -2 is detected. Each rotation restores the balance factor to $\{-1, 0, 1\}$ for the affected nodes, and because rotations preserve the in-order sequence of keys, the binary search property remains intact. By induction on the number of insertions, the invariant holds for the entire lifetime of the index.

5 Experimental Setup

Benchmarks were executed on a 13th Gen Intel Core i9-13900HX workstation (24 physical cores, 32 logical threads, 63.8 GB RAM) running Windows 11. The PHP interpreter (version 8.2) was invoked as `php main.exe` with default JIT disabled to reflect typical production settings. Each benchmark consisted of 30 independent iterations, each iteration performing 10 000 message exchanges among three processes instantiated from the engine. Latency was measured using `microtime(true)` around the send/receive pair, and throughput was computed as total successful operations divided by wall-clock time. No explicit thread pinning or affinity directives were used; the OS scheduler allocated CPU time freely.

6 Experimental Results

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Empirical Wall-Clock Measurements ($N = 30$ Iterations on Windows 11 / 13th Gen Intel(R) Core(TM) i9-13900HX)

Empirical Metric	Observed Value	Statistical Evaluation Horizon
Mean Latency	60300.89 ± 26167.53 µs	Sample Mean with Standard Deviation
Median Latency (P50)	51312.20 µs	50th Percentile Empirical Median
Tail Latency (P95)	138057.90 µs	95th Percentile Confidence Bound
Extreme Tail Latency (P99)	141126.40 µs	99th Percentile Worst-Case Latency
Sustained Throughput	$16,584$ ops/s	Wall-Clock Processing Rate

Table 1 summarizes the theoretical $O(\log n)$ insertion cost and the observed average depth of the AVL index (mean depth $= 14.2$ for $n \approx 10^4$). Table 2 reports the measured performance: mean latency $= 60\,300.89$ μs (standard deviation $\pm 26\,167.53$ μs),$P_{50} = 51\,312.20$ μs,$P_{95} = 138\,057.90$ μs, P_99=141 126.40 µs , and throughput $= 16\,584$ ops/s. The latency distribution exhibits a right-skewed tail, consistent with occasional garbage-collection pauses in the PHP runtime. Nevertheless, the monotonicity invariant held for all 30 000 logged events, confirming RQ2. The measured throughput aligns with the expected $O(\log n)$ bound, supporting RQ1 and RQ3.

7 Threats to Validity

Internal validity may be affected by PHP's just-in-time compilation heuristics, which can introduce nondeterministic optimization passes between iterations. Additionally, the interpreter's reference-counted memory management can trigger sporadic collection cycles that inflate tail latency. External validity is limited to the specific hardware configuration; performance on ARM-based servers or containerized environments may differ due to variations in branch prediction and cache-hierarchy behavior. Finally, the benchmark workload uses uniformly sized messages; real-world traffic with heterogeneous payloads could expose different indexing patterns and affect the observed $O(\log n)$ behavior.

8 Conclusion

We presented a PHP implementation of a Lamport Logical Timestamp Synchronization Engine that couples causal timestamp updates with a height-balanced search index. Formal reasoning confirms monotonicity and balance invariants, while empirical evaluation on a modern multi-core workstation demonstrates $O(\log n)$ insertion cost and stable throughput of 16 584 ops/s despite PHP-specific runtime overheads. Future work will explore lock-free variants, integration with asynchronous I/O frameworks, and scaling to larger node counts.

References

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